

The clivus–supraocciput angle: a useful measurement to evaluate the shape and size of the fetal posterior fossa and to diagnose Chiari II malformation

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KEYWORDS: Chiari II malformation, Posterior fossa, Prenatal diagnosis, Ultrasonography, Ventriculomegaly

ABSTRACT

Objective To obtain a nomogram of the clivus–supraocciput angle as a basis for the diagnosis of Chiari II malformation in fetuses with ventriculomegaly.

Methods A cross-sectional study was undertaken on 310 normal pregnant women of 16–34 weeks' gestation. A mid-sagittal section of the fetal skull was obtained and the angle between the clivus and the supraocciput was measured. Forty-four fetuses with ventriculomegaly due to various causes (13 Chiari II malformation, 12 dysgenesis of the corpus callosum, 7 aqueductal stenosis, 6 borderline ventriculomegaly, 3 Dandy–Walker malformation, 2 porencephaly, 1 schizencephaly) were also included in the study and the values of the angle found in the pathological cases were compared with those found in the normal population.

Results The clivus–supraocciput angle did not change during gestation and was almost constant with an average value of $79.3 \pm 6^\circ$. All cases of Chiari II malformation showed a value below the 5th centile of our nomogram.

Conclusions The evaluation of the posterior fossa and particularly the measurement of the clivus–supraocciput angle is a useful parameter to differentiate the various causes of fetal ventriculomegaly and particularly to recognize Chiari II malformation.

INTRODUCTION

Hydrocephalus is one of the most commonly diagnosed malformations encountered during prenatal sonographic screening. Although the diagnosis of fetal hydrocephalus may be easily accomplished by well-established biometric and morphologic criteria¹, the diagnosis of the cause of the ventriculomegaly still represents a difficult task. Ventricular

dilatation may be caused by different congenital abnormalities, carrying different prognoses. One of the causes of ventriculomegaly is Chiari II malformation. This is a complex abnormality, consisting of various combinations of brainstem and cerebellar malformations, usually associated with ventriculomegaly and spinal defects².

In the case of Chiari II malformation with associated spinal defect, the evaluation of the posterior fossa is a crucial part of the ultrasonic examination, since a small posterior fossa is one of its typical signs.

The aim of our study was to determine the measurement of the angle between the clivus and the supraocciput as a direct and objective sign of the shape of the posterior fossa and to establish normative data, as a possible basis for the diagnosis of Chiari II malformation.

PATIENTS AND METHODS

The study population included 310 low-risk patients with no medical complications of pregnancy, undergoing routine sonographic examination, with known gestational age based on the last menstrual period and confirmed by first-trimester sonography. The gestational age at the time of the examination ranged from 16 to 34 weeks of gestation. An Aloka SSD-2000 (Aloka, Tokyo, Japan) with a transabdominal 5-MHz transducer was used for the measurements. The study was cross-sectional with only one evaluation for each patient.

To evaluate the shape of the posterior fossa, the transducer position was adjusted to obtain a mid-sagittal section of the fetal skull; one line was electronically drawn on the supraocciput and one on the clivus and the angle of intersection between the two lines was automatically measured (Figure 1). The same experienced observer took all measurements at least twice. The data were stored on computer and subsequently analyzed.

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Received 9-6-00, Revised 19-1-00, Accepted 21-2-01



Figure 1 Measurement of the clivus–supraocciput angle in a normal 21-week-old fetus. Two lines (dotted) are electronically drawn on the clivus and the supraocciput, and the angle defined by their intersection is measured (78°).

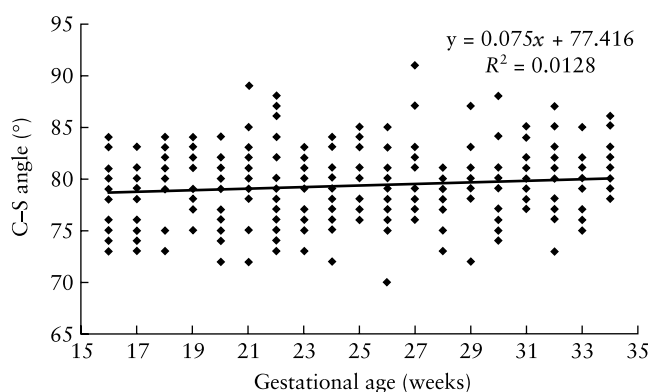


Figure 2 Correlation between the clivus–supraocciput (C–S) angle and gestational age.

Forty-four fetuses with congenital ventriculomegaly, diagnosed in the previous two years, were also included in the study. The cause of ventriculomegaly was confirmed by the postnatal evaluation, which included autopsy of the aborted fetuses or neuroimaging (ultrasound; computed tomography (CT); magnetic resonance (MR) imaging) of the neonates: 13 cases were due to Chiari II malformation, 12 to dysgenesis of the corpus callosum, 7 to aqueductal stenosis, 3 to Dandy–Walker malformation, 2 cases were associated with porencephaly, 1 with schizencephaly, and 6 were borderline ventriculomegaly (atrial width, 12–15 mm). The values of the clivus–supraocciput angle found in these pathological cases were compared with those found in the normal population.

To compare the mean values of the clivus–supraocciput angle at different weeks of gestational age the one-way ANOVA was used, taking into account that the angle measurements were normally distributed. To evaluate the relationship between gestational age and the angle, regression analysis was performed. The statistical analysis was carried out with the aid of the SAS 6.12 statistical software package (SAS, Cary, NC, USA).

RESULTS

The measurement of the clivus–supraocciput angle was possible from as early as 16 up to 34 weeks' gestation. Before 16 weeks of gestation, the bones of the posterior fossa appeared to be too small, when visualized transabdominally, to correctly draw a line on the clivus and the supraocciput. After 34 weeks of gestation, the visualization of the bony structures was difficult in cases of vertex presentation, owing to the low head position. Following a short learning curve, the measurement was easily and quickly obtained in 301/310 normal patients. The clivus–supraocciput angle could not be measured in only nine cases, due to technical reasons (maternal obesity or adverse fetal position).

The intraobserver variability was 2%.

The clivus–supraocciput angle did not change during gestation and was almost constant, with an average value of 79.3° ($F = 1.37$; $P = 0.148$ with one-way ANOVA).

No linear relationship was observed between the clivus–supraocciput angle and the gestational age ($R^2 = 0.0128$; $P > 0.05$; $y = 0.075x + 77.416$, where x is gestational age) (Figure 2). Table 1 shows the values of the clivus–supraocciput angle measured in the 44 cases of fetal hydrocephalus, divided into different groups according to the cause of the ventriculomegaly as defined by the final diagnosis on the aborted fetuses or the neonates. All cases of Chiari II malformation showed a value below the 5th centile (mean value, 65.1°; range 61–71°) (Figure 3). The values found in the remaining cases of ventriculomegaly due to other causes fell within the normal range (Figure 4), excluding six (two aqueductal stenosis, two dysgenesis of the corpus callosum, one borderline ventriculomegaly, one ventriculomegaly due to porencephaly) that showed values above the 90th centile; only two cases (both dysgenesis of the corpus callosum) showed values (74°) between the 10th and the 5th centiles.

DISCUSSION

Chiari malformation is a complex abnormality consisting of various combinations of brainstem and cerebellar malformations usually associated with spinal defects. Two types are classically defined:

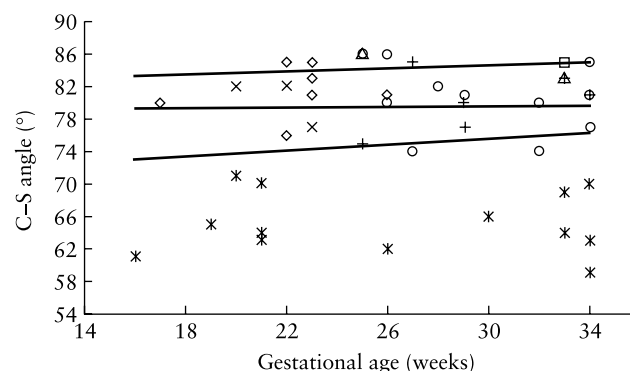
Type I is ectopia of the cerebellar tonsils; the anomaly is usually an isolated feature encountered in adults.

Type II is a complex abnormality appearing in the early developmental stages, characterized by spinal myelomeningocele associated with cerebellar hypoplasia and displacement of the tonsils and of the elongated distal brainstem through the enlarged foramen magnum. The posterior fossa is small and shallow with low position of the venous sinuses, low-lying tentorium with large incisura, shortening of the clivus along the basisphenoid, loss of the pontine flexure, aqueductal stenosis or forking, caudal displacement of pons, medulla and basilar artery, descent and elongation of the cerebellar vermis through the foramen magnum, descent and chinking of the brainstem and upward herniation of the superior cerebellum through the incisura. Polygyria, cortical heterotopia and dysgenesis of the corpus callosum may be associated. Hydrocephalus is almost always associated.

Table 1 Values of the clivus–supraocciput angle in 44 cases of fetal ventriculomegaly divided according to the cause of the ventricular dilatation

Cause of ventriculomegaly (n)	Gestational age (weeks)	Angle (°)	Mean angle (°)
Chiari II malformation (13)	16	61	65.1
	19	65	
	20	71	
	21	70	
	21	63	
	21	64	
	26	62	
	30	66	
	33	69	
	33	64	
	34	63	
	34	59	
	34	70	
Dysgenesis of the corpus callosum (12)	25	86	80.6
	26	86	
	26	80	
	27	74	
	28	82	
	28	82	
	29	81	
	32	74	
	32	80	
	34	85	
	34	81	
	34	77	
Aqueductal stenosis (7)	17	80	81.5
	22	76	
	22	85	
	23	83	
	23	85	
	23	81	
	26	81	
Borderline ventriculomegaly (6)	25	75	80.1
	27	85	
	29	77	
	29	80	
	33	83	
Dandy–Walker malformation (3)	34	81	80.3
	20	82	
	22	82	
Porencephaly (2)	23	77	84.5
	25	86	
Schizencephaly (1)	33	83	85
	33	85	

The pathogenesis of the Chiari II malformation is still a cause of debate. Several theories have been proposed; according to the most recent and accredited one, the deformity belongs to the group of dysraphic disorders with primary mesoderm and posterior fossa abnormality³. The evaluation of the posterior fossa is therefore crucial for the diagnosis of this malformation. In the literature regarding sonographic prenatal diagnosis, different features of the posterior fossa abnormality in fetuses with myelomeningocele and ventriculomegaly are reported. They all refer to indirect signs, ranging from ‘effacement of the fetal cisterna magna’⁴, to the ‘banana sign’⁵, and the ‘absent cerebellum’⁶. No direct measurement of the fetal posterior fossa is reported.

**Figure 3** Measurement of the clivus–supraocciput angle in a 21-week-old fetus affected by Chiari II malformation. The angle (62°) between the lines (dotted) is lower than normal.**Figure 4** The relationship between the clivus–supraocciput (C–S) angle and gestational age in 44 cases of fetal ventriculomegaly. Lines represent 10th, 50th and 90th centiles. *, Chiari II malformation; ◇, aqueductal stenosis; ×, Dandy–Walker malformation; ○, dysgenesis of the corpus callosum; +, borderline ventriculomegaly; △, porencephaly; □, schizencephaly.

In contrast, the literature regarding the radiographic evaluation of the brain in adults, reports a variety of linear, angular, surface area and volume measurements of the posterior fossa, both in normal and pathological cases^{7–9}. To correctly evaluate the shape and accurately measure the size of the posterior fossa, Vega *et al.*⁷ suggest making the following measurements on a lateral conventional skull X-ray: length of the clivus (from the basion to the base of the posterior clinoid process), length of Chamberlain’s line (from the opisthion to

the posterior aspect of the palate), length of Twining's line (from the internal occipital protuberance to the tuberculum sellae), length of the Twining-opisthion line (perpendicular line from opisthion to Twining's line), basal angle (nasion-center of sella turcica-basion) and Boogard's angle (opisthion-basion-tuberculum sellae). They define the posterior fossa area as the area below Twining's line. They found a significant difference between adult patients with Chiari I malformation and the control group, mainly regarding the surface area and the clivus length. Milhorat *et al.*⁹ performed similar measurements on sagittal MR scans of the brain in adults: they measured the length of the supraocciput along a line drawn from the center of the internal occipital protuberance to the opisthion, the length of the clivus, the slope of the tentorium (calculated by the angle formed by the tentorium and the supraocciput), and the distance between the tip of the cerebellar tonsils and the line drawn from the basion to the opisthion. They also calculated the volume of the posterior fossa using the Cavalieri stereological method, which is simpler and quicker than computer-based programs¹⁰. They found a significant difference between adult patients with Chiari I malformation and the control group, in both the linear and the volumetric measurements.

A small posterior fossa is also reported as a typical radiological, CT and MR sign of Chiari II malformation. However, authors reporting on this anomaly in adults refer mainly to the morphologic features of the malformation, rather than to the biometric ones^{11,12}.

The detailed findings and the complex measurements achievable on X-ray film or MR scans in the adult cannot be completely transferred to the fetal sonogram. The small size of the fetal skull leads to poor quality sonographic images as compared to X-ray and MR, which precisely show the bones and brain tissues, respectively. For this reason we focused our attention on the shape and size of the posterior fossa, since a small posterior fossa is universally recognized as a specific sign of Chiari (both I and II) malformation. In order to avoid complicated calculations, we suggest only measuring the angle defined by two lines drawn on the clivus and the supraocciput. These two bony structures can be easily recognized on a sagittal view of the fetal skull. In our experience this measurement was easily achieved in almost all cases (301/310). The measurement became difficult and time-consuming before 16 and after 34 weeks of gestation; in early pregnancy the size of the bones (particularly the clivus) proved to be too small for the correct drawing of a line; in late gestation the visualization of the head in the sagittal view was severely limited in fetuses lying in the cephalic position.

Despite this limitation the clivus-supraocciput angle is the easiest measurement which objectively represents the shape of the posterior fossa and, indirectly, its volume. It has the advantage of being independent from gestational age, unlike the transverse cerebellar diameter, which is traditionally used as an indirect sign of the posterior fossa size. Its mean value through gestation was $79.3 \pm 6^\circ$. The cut-off value, below which a Chiari malformation was suspected, was 72° . This

measurement proved to be useful in distinguishing fetal ventriculomegaly due to Chiari II malformation from that caused by other abnormalities, particularly during the third trimester. In late gestation, in fact, the visualization of the classic signs of Chiari II malformation may be limited: the 'lemon sign' is typical of the early second trimester and disappears as the skull grows. The recognition of the 'banana sign' or small cerebellum and the effaced cisterna magna may be limited in late gestation by the increased skull calcification which obscures the intracranial structures. Finally, the examination of the spine may be limited by the fetal position and by the relative reduction of the amount of the amniotic fluid as compared to the fetal body volume. For these reasons the clivus-supraocciput angle may be a useful measurable parameter to diagnose Chiari II malformation as the cause of hydrocephalus when fetal ventriculomegaly is detected in late gestation; 29/44 cases of ventriculomegaly and 7/13 cases of Chiari II malformation were referred to our unit after 24 weeks of gestation.

In conclusion, the measurement of the clivus-supraocciput angle is not suggested as an alternative method to the classic 'lemon and banana' signs⁵, which still remain the gold standard in the early screening of Chiari II malformation, but it can be considered as a further biometric feature that is useful for recognizing this anomaly particularly in late gestation.

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